Internet of Bodies – IoB

https://storagecityplatform.com/English/blog-detail/Internet%20of%20Bodies%20(IoB)
https://www.forbes.com/sites/bernardmarr/2019/12/06/what-is-the-internet-of-bodies-and-how-is-it-changing-our-world/?sh=231386b68b7a
https://www.datadriveninvestor.com/2020/03/02/internet-of-bodies-a-key-notch-in-medical-sustenance-yet-another-wave-of-technological-swindle/

หลายคนอาจจะเคยชินกับ IoT ที่ทุกสิ่งสามารถควบคุมและเชื่อมต่อผ่านอินเทอร์เน็ตได้ แต่มีอีกอย่างหนึ่งที่คล้ายกันและใกล้ตัวเรามาก ๆ นั่นคือ IoB หรือ Internet of Bodies

IoB ในปัจจุบันที่เราสามารถเห็นกันได้ง่าย ๆ เช่น เครื่องกระตุ้นหัวใจขนาดเล็กที่ฝังไว้ในช่องท้องหรือหน้าอก เพื่อช่วยให้ผู้ป่วยการควบคุมจังหวะการเต้นของหัวใจ โดยใช้แรงกระตุ้นจากกระแสไฟฟ้า หรือ Smart Watch ที่ใช้เก็บข้อมูลต่าง ๆ ในร่างกาย เช่น อุณหภูมิ อัตราเต้นของหัวใจ และอื่น ๆ เพื่อมาเป็นผลวิเคราะห์ทางสุขภาพ

นอกจากนี้ยังมีกรณีที่ IoB ได้ถูกนำไปใช้ในธุรกิจ เช่น การฝังชิปขนาดเล็กเอาไว้ในร่างกาย ทำให้เราเดินเข้าบ้านหรืออาคารต่าง ๆ ได้โดยไม่ต้องใช้บัตรหรือกุญแจ รวมไปถึงการซื้อของต่าง ๆ ก็สามารถซื้อได้เลย โดยไม่ต้องจ่ายเงินหรือใช้บัตรรูดแต่อย่างใด

ในปัจจุบัน แม้ IoB จะดูล้ำสมัยและมีประโยชน์ต่อการดำเนินชีวิต แต่ IoB ยังคงมีประเด็นในเรื่องความเป็นส่วนตัว และด้านจริยธรรมต่าง ๆ ซึ่งเป็นโจทย์ที่ท้าทายสำหรับนักพัฒนา ว่าจะแก้ไขปัญหาพวกนี้ไม่ให้เกิดผลกระทบได้อย่างไรในอนาคต

?Internet of Bodies (IoB) คือ การเชื่อมต่อระบบจัดเก็บและวิเคราะห์ข้อมูลเข้ากับร่างกายของเราผ่านอุปกรณ์ต่าง ๆ ที่ทำหน้าที่รับส่งข้อมูลของร่างกายเราส่งไปยังระบบผ่านเครือข่ายอินเทอร์เน็ต คล้ายกับแนวคิด Internet of Thing หรืออินเทอร์เน็ตของสรรพสิ่ง ข้อมูลจากร่างกายเราจะผ่านการวิเคราะห์โดยระบบ เพื่อดูว่ามีความผิดปกติใด ๆ ที่ต้องได้รับการแก้ไขหรือไม่ ซึ่งโดยส่วนใหญ่แล้ว IoB มักจะใช้เพื่อการติดตามข้อมูลด้านสุขภาพของเรานั่นเอง

การติดตั้งอุปกรณ์ที่ใช้กับร่างกายเรานั้น มีหลายแบบ ตัวอย่างการติดอยู่บนร่างกาย เช่น สมาร์ทวอช การใส่ไว้ในร่างกายอย่างเครื่องกระตุ้นหัวใจ หรือกระทั่งฝังไว้ในร่างกาย

What is the Internet of Bodies (IoB)?

When the Internet of Things (IoT) connects with your body, the result is the Internet of Bodies (IoB). The Internet of Bodies (IoB) is an extension of the IoT and basically connects the human body to a network through devices that are ingested, implanted, or connected to the body in some way. Once connected, data can be exchanged, and the body and device can be remotely monitored and controlled.

There are three generations of Internet of Bodies that include:

- **Body external:** These are wearable devices such as Apple Watches or Fitbits that can monitor our health.
- **Body internal:** These include pacemakers, cochlear implants, and digital pills that go inside our bodies to monitor or control various aspects of our health.
- **Body embedded:** The third generation of the Internet of Bodies is embedded technology where technology and the human body are melded together and have a real-time connection to a remote machine.

What Is The Internet Of Bodies? And How Is It Changing Our World?

Bernard MarrContributor
Enterprise Tech

Have you heard the term the Internet of Bodies (IoB)? That may conjure up a few thoughts that have nothing to do with the true nature of the term, but it's about using the human body as the latest data platform. At first, this concept seems quite creepy, but then when you realize the possibilities it creates, it becomes quite exciting. Here we explore what the Internet of Bodies is, some examples in use today, and a few of the challenges it presents.

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ADOBE STOCK

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Progress in wireless connectivity, materials, and tech innovation is allowing implantable medical devices (IMD) to scale and be viable in many applications.

Examples of Internet of Bodies Devices in Use or Development

The most recognized example of Internet of Bodies is a defibrillator or pacemaker, a small device placed in the abdomen or chest to help patients with heart conditions control abnormal heart rhythms with electrical impulses. In 2013, former United States Vice President Dick Cheney got his WiFi-connected defibrillator replaced with one without WiFi capacity. It was feared that he could be assassinated by electric shock if a rogue agent hacked the device.

A "smart pill" is another IoB device. These pills have edible electronic sensors and computer chips in them. Once swallowed, these digital pills can collect data from our organs and then send it to a remote device connected to the internet. The first digital chemotherapy pill is now in use that combines chemotherapy drugs with a sensor that captures, records, and shares information with healthcare providers (with the patient's consent) regarding the drug dosage and time, plus other data on rest and activity, heart rate and more.

"Smart contact lenses" are being developed that integrate sensors and chips that can monitor health diagnostics based on information from the eye and eye fluid. One smart contact lens in development aims to monitor glucose levels that will hopefully allow diabetics to monitor their glucose levels without repeated pinpricks throughout the day.

Taking it up a notch is the Brain Computer Interface (BCI), where a person's brain is actually merged with an external device for monitoring and controlling in real-time. The ultimate goal is to help restore function to individuals with disabilities by using brain signals rather than conventional neuromuscular pathways.

But not all Internet of Bodies use cases are for healthcare reasons. Bioengineering company, Biohax has embedded chips in more than 4,000 people primarily for convenience. In one widely reported example, 50 employees of Three Square Market agreed to have an RFID microchip the size of a large grain of rice (similar to what's embedded in pets to be able to identify and locate them when they are lost) implanted. This chip allows these employees to gain access to the building without a key, pay for items with a wave of their hand at the vending machine by deducting the amount immediately from their account rather than use money and log onto their computers.

Challenges Faced by Internet of Bodies Technology

The situation of U.S. Vice President Cheney getting a defibrillator not connected to WiFi for security reasons illustrates one of the biggest challenges faced by Internet of Bodies technology—how to secure the devices and information they collect and transmit. Nearly half a million pacemakers were recalled in 2017 by the U.S. Food and Drug Administration over security issues requiring a firmware update. The security challenges faced by Internet of Bodies tech are similar to what plagues Internet of Things generally, but there can be life and death consequences when IoB devices are involved. Additionally, IoB devices create another cyber security challenge that will need to be safeguarded from hackers.

Privacy is also of paramount concern. Questions about who can access the data and for what purpose need answers. For example, a device that monitors health diagnostics could also track unhealthy behaviors. Will health insurance companies be able to deny coverage when a customer's IoB device reports their behavior? A cochlear implant could restore hearing, but it might also record all audio in a person's environment. Will that data remain private?

As Internet of Bodies tech continues to grow, regulatory and legal issues will have to be resolved and policies built around the proper use of the technology.

Healthcare

Internet of Things

Technology

Internet of Bodies (IoB)

By Adam Tabriz, MDMarch 2, 2020

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A key notch in medical sustenance; yet another wave of technological swindle

The Internet

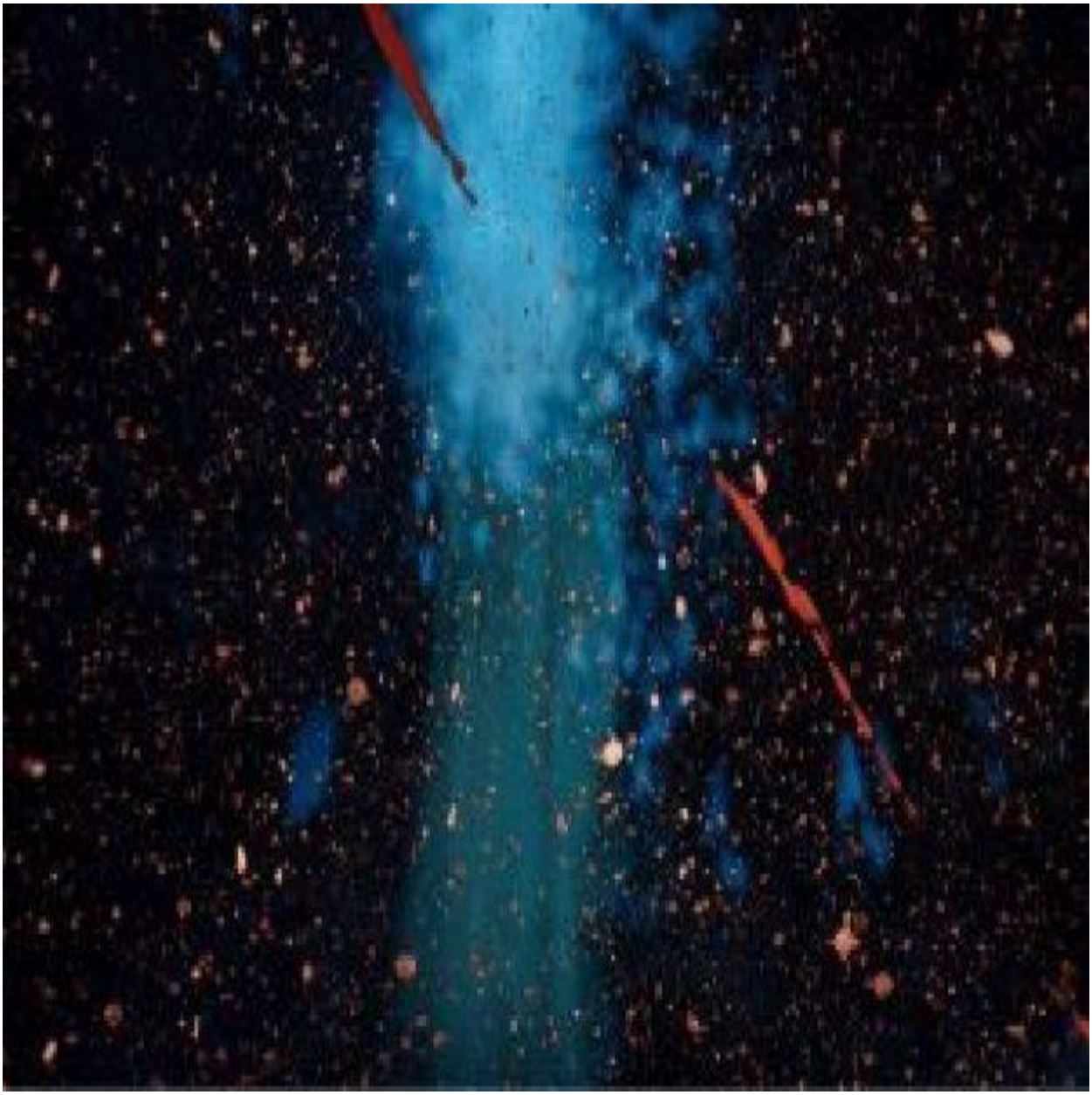
Internet and the Internet of Things (IoT) are a couple of the most revolutionary inventions of human history. Its primary origin goes back to the "United States Advanced Research Projects Agency" (DARPA). It was deployed for the first time in October 1969. Then it was referred to as ARPANET. Following that, the

contemporary Internet as" World Wide Web" (WWW) was built by CERN in Switzerland in 1990 by a British (UK) scientist named Tim Berners-Lee.

Today, the Internet is the biggest and the most robust world-wide communication system between computers. The modern Internet comprises millions of smaller domestic, academic, business, and government networks. Together they transmit a myriad of various sorts of data.

The Internet or, in short, the 'net' is the largest service network. The World Wide Web is used by billions of people all over the world. Today, the Internet has rooted in the day to day lives of ordinary people from providing them with tools to pay money and obtain services to automate daily chores. Some services on the Internet costs naught to use, but the rest may be accessed for a fee.

Sometimes people who offer free services as a form of software as a service (SAAS) use advertising or media coverage to make money. Latter has created a particular controversy with regards to censorship and freedom of speech on the Internet. While we are decades after the initial deployment of the Internet, we face overwhelming challenges concerning the net-neutrality of internet and SAAS providers.



The evolution of digital communication and the Internet

With the evolution of the Internet, communication has evolved flawlessly. Parallel to that, it has also become dangerous. Because users often send private information, therefore predisposing them to third party extortion of the transmitted data. Any bystander can also use the Internet to spread out fables or embezzled enigmas or dangerously wrong advice. Some of the social media sites like Facebook have chronologically had some dilemmas with privacy arenas. Some websites may con users into downloading harmful viruses aimed to harm a computer or spyware to hijack its user's private information. In the world-wide-web chat rooms, people have the potential to stalk or abuse other users. Also, the Internet comprises contents that users may

find offensive, such as pornography. Likewise, Criminals may well kidnap people's confidential information or fool people into transferring them money.

The era of data rush and Value of data, and Internet

With over two decades into the Internet era, as well as the personal success of streamlining data exchange, so became the Value of data. The 21st- century is the epoch of monetizing on data, resembling the 19th-century American gold rush. The servers of the tech industry that provide SAAS solutions serve as" silos" for billions of dollars with public data. The centralized nature of the data stored has merely placed the modern gold mines at giant corporate fingertips. Today, The liability of people to the particular collective action of the corporations is actual, as they curb public perception through social media and full access to sensitive information. Internet is the vehicle for data storage, analysis, modification, and most of all, the monetization. Cyber data are raw and unlimited. It is hence irresistible to profiteering entities.

Internet of things (IoT)

Before the rollout of the modern Internet in the 1990s, the concept of a network of smart devices was already a topic of discussion. First introduced in 1982 was the very first prototype used that modified the Coke vending machine at Carnegie Mellon University. The latest marked the first Internet-connected device. The appliance was able to report its inventory, so the temperature of the drinks.

The Internet of Things (IoT) is a network of interrelated computing tools, mechanical and digital machines, objects, animals, or people. Every node or thing within the system holds unique identifiers (UIDs), thus the ability to disseminate data over the web without compelling human-to-human or human-to-computer dealings.

The definition of the Internet of Things (IoT)

The meaning of the Internet of Things has perpetually unfolded. Over decades it has emerged into various technologies, real-time analytics, machine learning, commodity sensors, and ingrained systems. In layperson's terms, IoT technology is the "smart home," covering devices and appliances. They benefit one or more widespread ecosystems and can be controlled via devices attributed to that ecosystem, such as smartphones and smart speakers. Traditional fields of embedded systems are wireless sensor networks, control systems, automation (including home and building automation).

Parallel to all the benefits associated with the use of IoT, there is an abundance of significant considerations about threats in the development of IoT. That applies particularly in the sectors of privacy and security, necessitating industry, and governmental actions to address. For instance, The IoT suffers from protocol fragmentation and a lack of technical standards. Meaning, There seems to be a variety of IoT devices in terms of hardware variations and differences in the incorporated software content. To further elaborate with an example, one can note the wireless connectivity for IoT devices. Technologies using Bluetooth, Zigbee, Z-Wave, LoRa, NB-IoT, Cat M1, as well as custom proprietary radios – each possess its benefits and flaws; and unique permission ecosystem.

IoT's unstructured computing essence is problematic for security. The patches of bugs found in the core operating system often do not reach users of older and lower-price devices. Some researchers assert that the failure of vendors to support older equipment with patches and updates leaves most of the active Android devices vulnerable to hacking.

The Internet of Things (IoT)and privacy

Questions over privacy have led many to consider the likelihood that big data infrastructures such as the Internet of things and data mining are innately incompatible with privacy. Peter-Paul Verbeek, a professor of philosophy of technology at the University of Twente, Netherlands, jots down that technology impacts human moral judgment, which in turn affects social mechanism, privacy, and sovereignty. In his writing, Verbeek warns against viewing technology merely as a personal tool. He advocates instead of considering it as an active agent. It involves every aspect of human life, from simple control of the doorbell and home security systems to patient overhaul and elderly care.

The Internet of things is exceptionally scalable, incorporated into more profound aspects of our lives, and is becoming more and more sophisticated, primarily through exploiting deep learning technology. The liberal access to personal information makes this trend even more complicated.

Classification of Internet of things (IoT)

As highlighted earlier, the IoT is a general term for a system of connectivity between a vast number of devices or "things" amid the enormous amount of domains. For example, the Internet of Medical things (IoMT) or Internet of bodies (IoB).

The Internet of Medical Things, (also called the Internet of health things), is an application of the IoT remaining medical and health domain. The IoMT, or "Smart Healthcare," is a technology utilized for constructing a digitized healthcare system by engaging available medical resources and healthcare services.

Role of Sensors in the Internet of Things (IoT)

Most of us are familiar with technologies such as blood pressure, heart rate monitors, advanced devices capable of monitoring specialized implants. Few other well-known technologies include pacemakers, Fitbit electronic wristbands, or advanced hearing aids. The sensors in the IoT network exist to facilitate remote health monitoring and emergency warning networks. The sensors within IoMT, in turn, establish a network of intelligent feelers. They can collect, process, transfer, and analyze valuable information in various settings, such as connecting in-home monitoring devices to hospital-based systems. Different accessories that motivate healthy living, such as connected scales or wearable and tools also can complement IoT. Applications for point-of-care medical diagnostics, where portability and low system-complexity is essential.

Nevertheless, Practical supervision of devices to promote credible, secure, and high-quality applications is challenging parallel to the increasing scale of the system. Therefore, it has been proposed that, as one of the critical cornerstones of IoT device management, automatic cross-device classification aiming to spot the semantic type of a machine by analyzing its network traffic. This has the potential to support an extensive range of features. It includes enhanced security by requiring the proper rules for restricting the communications of distinct types of devices or context-awareness of IoT applications.

Internet of Bodies (IoB)

Internet of Bodies relates to devices, placed on or implanted into the human body to carry on a continual dialogue among a variety of users via the Internet. No doubt; The Internet of bodies enables people who can live better and longer with medical conditions such as diabetes by enabling us to monitor blood glucose closely and control body functions, therapies, and functions, using a simple smartphone app. For instance, a new generation of insulin pumps connects to an app that communicates wirelessly with the pumps while sending data to the cloud for other diagnostic and therapeutic uses. Seemly, such devices that might be more accurately refer to as systems are creating new concerns about human elements.

Related Article: Deep learning Gamification and Artificial Intelligence as a modern learning tool; or an agent of indoctrination

Ingestible sensors and Internet of Bodies (IoB)

Likewise- the modern and sophisticated “ingestible sensors” are starting to emerge into the healthcare market. It will potentially be delivered in the form of packing the electronics technology into an ingestible capsule with a semi-permeable coating. Once swallowed, they assess gut climate as the sensor moves along, pinging data to the smartphone. Along with technological advancement, users must maintain awareness of what their IoB does. And they must prepare ready to step in when needed. Thus users need to sustain a reasonable level of situational insight and be prepared to exercise control.

It has been postulated; with the continuous advancement of the big data-powered Internet of Things (IoT) into the Internet of Bodies (IoB), the integrity of the system of human bodies relies at least partly on the Internet and related technologies, such as artificial intelligence. This will predispose human body devices to

the same levels of security flaws that potentially afflict IoT. Yet, unlike most IoT- IoB will directly and physically harm the human body. The Internet of Bodies will hail the arrival of corporate software liability and a new legal and policy battle over the integrity of the human body and mind.

Real-time seamless data flow artificial intelligence and risk

In addition to functional and privacy risks associated with IoT and IoB, there is another substantial point to note. The indiscriminate sharing of personal data with 3rd party entities and corporations such as insurance companies increasing apparent risk. In connection with Artificial intelligence algorithms formulated to target particular objectives, the IoB will expose patients with preexisting medical conditions to the spike of high premiums without patients' awareness.

According to Business Insider, it is estimated that there will be over 64 billion IoT appliances in use by 2025, up from about 9 billion in 2017. The IoT devices will generate a lot of data that needs to be collected and mined for execution in one way or another. Artificial Intelligence will be used to process and analyze those data. Dissociation of the healthcare community from the technology has created a vacuum of sparse algorithms validation and data ownership.

Corporate tactical mission strategic endeavor, and individual vulnerability

It is the tactical objective of corporations to occupy the IoT space. This way, they can facilitate secure and efficient harnessing public data to elicit functions that are driven by those missions. Today, corporations frequently pursue strategies to ensure a maximum fiscal gain. Large entities have lobbied their way to tightened competition as well as tight control over the market.

Historically corporations have relished the exorbitant of personhood, the collective influence of its stakeholders, money, and technology. Today, they are on the verge of exercising the ability to read the human mind, access their personal information without breaking a single law.

Wrong Steps in the wrong direction

The chronological review has demonstrated a trend of wrong steps in the wrong direction. The leaning points to; there is no reason why we should expect any different scenarios concerning the Internet of bodies. For example, Amazon Alexa has a track record of collected data via voice recognition technology from households without significant legal implications. Or The National Institute of Health's (NIH) focus of primarily genetic sequencing and expanding upon the genomic database will also flourish the availableness of information to other organizations and agencies. Yet, they have little or nothing to do with precision medicine.

Further, Controversies are related to feeding the Big-Data-Driven Stem Cell Science to other entities by Food and drug administration.

Millennial role and IoB

Generally speaking, Millennial is pro-technology. However, interestingly, studies suggest that the Millennial generation has some issues with the Internet of Things. The review is suggestive of a trend that presents a

challenge for utilities. Attempting to facilitate policies like demand response that can link to IoT products such as smart thermostats, air conditioners, or appliances are becoming a strenuous task.

According to a novel cohort study, young American adults 18 to 21 years are less likely to own an IoT product. But further research advocates the main reason for lower ownership of IoT devices among Millennials is that they are less likely to be homeowners. Therefore, they are not as expected in the market to buy IoT devices that can help manage energy usage. In fact, Millennials construct the heart of why the IoT exists and why it will endure. Because; the majority spend up to six hours of productive time daily online. More than two-thirds of them have smartphones. We must keep in mind that the tech industries are on a conquest to curb data and artificial intelligence. The Modern millennial perceptual experience of personalized and population healthcare is multifarious, hence is fueling their endeavor.

More than ever, The new generation feels technological evolution would be enough to ascertain bespoke healthcare; hence along with the same mindset, tailored medicine is perceived to be synonymous with genomics.

The Internet of bodies and its surprises

With the ever-emerging Internet of Things, humans have become somewhat comfortable with the fact that our tools and equipment know more about us than we identify about ourselves, and our smartphones are disseminating our every footstep. The IoT has even made it into the courtrooms. For example, it has been postulated that Amazon Echo's Alexa can be a perfect witness in the court. The abrupt momentum of technological transformation proceeds at a nearly exponential proportion. Meanwhile, it is having a tremendous influence on the public psyche and data privacy. Over the past decades, technological progressions in healthcare and medicine have incorporated upswings in AI to create an adventurous realm. Similar to hooking up digital appliances to the Internet, we are now trying to hook up human bodies to the Internet.

Dangers of the Internet of Bodies (IoB)

Just imagine a hacker attacks your body from across the world. Hence they are able to intentionally cause pacemaker malfunction or sending a signal to change your motor heart rhythm. And only believe that would make cyberwar of the future will look like?!

Given the potential for misuse of intimate biodata, it would not be startling that many top leaders and thinkers are now endorsing the passing of new legislation that can establish a new data privacy bracket for the Internet of Bodies. In fact, in April 2018, European nations led by the UK, France, and Estonia began working on a "European approach to AI" that would outline privacy rights and require companies and governments to put special safeguards into place.

Stride for technological transformation

The major problem is that the stride of technological transformation is much unexpected and fast than regulatory and healthcare communities are, and to afford to keep up. In such a scenario, regulators try to catch-up. At the same time, struggle to shut down all the loopholes and removing negative inducements from the marketplace. However, despite try, they are invariably one step behind corporations and the tech industry.

According to few sources, mainstream tech journals have been failing to emphasize warning signals of the "biohackers" threats. People desperately need to think about encrypting their DNA the same way that they encrypt other digital personal data.

Internet of Bodies (IoB) and technological mediation

Peter-Paul Verbeek is a Dutch philosopher of technology, chair of the philosophy department at the University of Twente, Netherlands, a member of the Dutch Council for the Humanities and chair of the Society for Philosophy and Technology. He is the proponent of the theory of technological mediation that builds on the post phenomenological approach, first stipulated by Don Ihde. Verbeek presents his method of technological intervention to systematically analyze the effect of technology on human attitude. And in terms of how the role, technology plays in human-world relations.

Human-technology-world relationship

In his original theory, Verbeek outlines several different human-technology-world relations:

1. Embodiment relations as technology does not call attention to itself but to aspects of the world given through it. (e.g., glasses)
2. Hermeneutic ties, where technology represents a distinct aspect of the world. (e.g., a thermometer)
3. Background relations in which technology forms the experiential context, going beyond conscious experience. (e.g., room temperature through a central heating system)

Alterity relations: in which technology presents itself as quasi other to the subject (e.g., an ATM)

5. Cyborg relations: in which technology merges with the human. (e.g., brain implants)
6. Immersion relations: in which technology forms an interactive context. (e.g., smart homes)
7. Augmentation relations: in which technology mediates and alters our experience of the world, e.g., Google Glass.

Verbeek's theory supports the underlying relationships between global culture and technology. It is the major driving factor behind millennial over-reliance and trust in fast pace technological revolution. That includes the Internet of bodies. But, the Internet of Things is evolving into a weapon against humanity; even through the corporate fundamental tactical mission is saving lives.

The path to Solution is straightforward; yet uneven

Data science and artificial intelligence are no fairy tales in the biomedical realm.

Hospitals and health insurance companies force Physicians to operate continuous data capture utilizing wearable medical devices. Still, classical healthcare heavily banks on increased precision sparsely on datasets. Parallel to that system, anticipates patients to get a comprehensive medical checkup once a month as opposed to continual monitoring of a handful of vital parameters.

Encouraging Physicians in the age of Dataism

Presently there is an ocean of dense low-quality data sets. That is why the most significant impact of data science will be encouraging physicians to extract clinically relevant information. While the tech industry is on

the driver seat en-route to mechanizing medical diagnosis, physicians are becoming alienated from the fast pace technologies. Yes- indeed, physicians need to be engaged, but on whose terms?!

That is the question!

The insurance industry, the tech industry, or a combination of the two?!

I recognize none!

Maintain Individual Control

It is just as vital to maintaining individual control over their own data. Of course, latter control merely would exist amid all strategies to ensure independent physician sovereignty and contribution to healthcare technology. It is also vital for administrations concomitantly to preserve transparency at every level from the research and development phase to execution and quality assurance.

Therefore, physicians' skills must be conformed to creating the right instrument that will help their domain. Only then will they be able to use them to achieve favorable results. The technologies can be used efficiently and safely only if we account for the decisions of the stakeholders involved in healthcare organizations.

Decentralization of Data

The first step is to decentralize data through blockchain while employing other data protection tactics. Unlike what is being conveyed by non-healthcare industries, Physicians must only use large volumes of data generated from continuous monitoring to complement their clinical judgment. Of course, we should learn to analyze multiparameter data, data visualization, and Understand AI workflow.

Understanding Statistical Significance

Last but not least must they must understand the statistical significance of clinical studies. And let us not forget technology can never replace human touch, merely because medicine is the science of infinite variability, even beyond the machine learning sphere.